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A patient information mining network for drug recommendation

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ABSTRACT

As an important task of natural language processing, medication recommendation aims to recommend medication combinations according to the electronic health record, which can also be regarded as a multi-label classification task. But patients often have multiple diseases simultaneously, and the model must consider drug-drug interactions (DDI) of medication combinations when recommending medications, making medication recommendation more difficult. There is little existing work to explore the changes in patient conditions. However, these changes may point to future trends in patient conditions that are critical for reducing DDI rates in recommended drug combinations. In this paper, we proposed the Patient Information Mining Network (PIMNet), which models the current core medications of patient by mining the temporal and spatial changes of patient medication order and patient condition vector, and allocates some auxiliary medications as the currently recommended medication combination. The experimental results show that the proposed model greatly reduces the recommended DDI of medications while achieving results no lower than the state-of-the-art results.

1. Introduction

Medication recommendation aim to provide patients with effective and safe medication combinations. Medication recommendation models can better help nurses understand patient's conditions and help doctors diagnose and treat patients more effectively, saving doctors valuable time from the fast-paced way of working in hospitals. Medication recommendations are usually based on a patient's electronic health records (EHR), which include a patient's diagnosis history and a doctor's prescription history. With the aid of the medication recommendation model, doctors can prescribe drugs faster in actual treatment. Patients may generally suffer from multiple diseases, so physicians often prescribe a combination of multiple drugs for each patient visit. However, due to the possible drug-drug interactions (DDI) between drugs, the drug recommendation model must consider both accuracy and DDI, which makes the task of drug recommendation more difficult.

In previous work, neural network-based medication recommendation can be divided into two categories: The first method does not consider the patient's historical information, the Instance-based methods, which makes recommendations based on the patient's current diagnostic record. This method fully mines the patient's current diagnostic vector, but the lack of the patient's historical information may lead to poor recommendation results. The second method is based on the patient longitudinal vector modeling, the Longitudinal method, which aims to capture the dependencies between patient medical history vectors and achieve better results. However, the DDI rate of the recommendation results given by the existing methods is too high, which greatly limits the application of the medication recommendation model.

At present, most modeling methods based on longitudinal vectors of patients are modeled from the perspective of medications, and rarely from the perspective of patient's conditions. However, in real life, if a doctor wants to prescribe a medication to a patient, he will prescribe the medication according to the patient's previous condition. Similarly, the patient's current condition is also largely related to the patient's previous condition. To illustrate the relationship between patient condition data, we take an example, as shown in Fig. 1. Identical diseases are marked in red in Fig. 1, and potentially related diseases are marked in purple. This patient had a total of 3 visits and found that each visit not only had some of the same diseases, but also some diseases related to the previously diagnosed diseases. Likewise, a doctor may prescribe the same drug for the same disease, and for a disease that has developed from a previous disease, a doctor may add new drugs to the previous drug. Therefore, ignoring the changing trend of the patient's condition and making the model unable to highlight the patient's core condition

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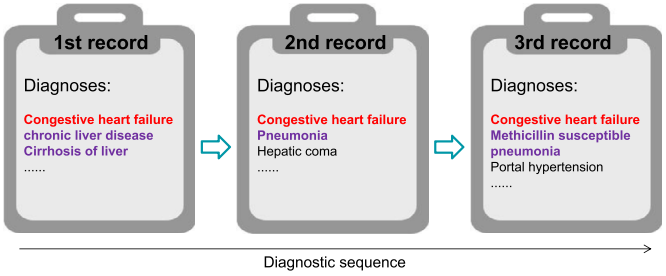


Fig. 1. An example of the relationship between a patient's condition history.

may lead to a high DDI of the model result. In some actual treatments, a higher DDI rate may be fatal.

We tried to explore the relationship between the diagnosis information of each visit of the same patient. We calculated the Jaccard coefficients and Ochiia coefficients between the disease information diagnosed each time as shown in Fig. 2. It can be found that the correlation is weak in terms of the disease names of the patients at each visit. However, we do know that there is a correlation between the patients' history. Therefore, we need better methods to mine the correlation between disease information and use this correlation to explore the impact of changes in patient's conditions on drug changes. In this paper, we proposed a model to capture the changes in patient's conditions, named Patient Information Mining Network (PIMNet). PIMNet can be mainly divided into the patient information mining module, the drug graph embedding module, and the holistic medication module. The patient information mining module predicts the patient's core conditions through learning the correlation between patient's historical condition and recommends the core medications. The core medications are the medications that play a key role in the treatment of the patient. This module greatly reduces the DDI rate of the recommended results while maintaining the Jaccard score and F1 score as much as possible. The drug graph embedding module is used to embed drug information. The holistic medication module only makes recommendations based on the patient's current health information, and the main task of this module is to correct. The main contributions of this paper are as follows:

- We propose a novel safer medication recommendation model, PIMNet, which can effectively reduce drug-drug interactions for recommended medication combinations while providing accurate recommendation results.
- Different from the vast majority of medication recommendation methods, we model from the perspective of patient information minings, which provides a new idea for medication recommendation methods.
- We conduct adequate experiments on the public dataset MIMIC-III, which demonstrate the effectiveness of PIMNet.

2. Related work

The task of medication recommendation is to give medication recommendation to patients based on the patient's medical information. According to the type and amount of information used by the method, existing methods can be divided into the rule-based methods, the instance-based methods, and the longitudinal methods.

2.1. Rule-based methods

Rule-based methods rely on rules designed by people to make recommendations. Typically, this approach requires a large number of physicians to design and is not very transferable. For example, [12] searches for a treatment option through a Markov decision-based strategy by formulating the treatment option as an if-then-else rule. Almirall et al. [1] and Gunlicks-Stoessel et al. [9] developed a new adaptive treatment strategy to recommend treatments. Chen et al. [3] uses chronic disease medical guidelines to build reasoning systems to make treatment recommendations.

2.2. Instance-based methods

Instance-based methods only use the current patient's health information and do not consider the patient's historical health information. Choi et al. [4] use two attention modules to model the patient vector and the drug vector and achieves better accuracy. Zhang et al. [24] uses reinforcement learning to fine-tune a decision model based on a sequence-to-sequence structure and joins an external dataset to reduce the rate of adverse drug interactions. The Instance-based methods may be less accurate due to lack of historical patient health information.

2.3. Longitudinal methods

Longitudinal methods utilize historical health information of patients, this method is more performant and more commonly used. Gong et al. [8] use medical knowledge graph to provide a safer recommendation. Bhoi et al. [2] use Graph Attention Network [20] to model drug co-occurrence and interaction relationships. Shang et al. [15] use the patient's historical health information to establish a Graph Augmented Memory Module, and use the Graph Convolutional Network [11] to model relations based on the medication EHR and DDI graphs. Yang et al. [22] proposes using multiple linear layers to extract changes in patient prescriptions to predict changes between current and previous prescriptions. Yang et al. [23] proposes a bimolecular graph encoder to use drug molecular structure to embed drug information and achieve lower DDI and better accuracy. Wu et al. [21] proposes a novel copy-or-predict mechanism to generate the set of medicines and achieve higher accuracy.

3. Method

To mine the relationship between patient's historical condition information, we propose a new medication recommendation model, named PIMNet. We will first define the notation and problems required by the model, and then present the overall outline and detailed structure of the model.

3.1. Definition

Definition 1 *(Electrical Health Records (EHR))*. A patient's EHR data includes detailed information about each visit, which is stored as a medical code and the longitudinal vectors of the patient. Let $X_i = \{x_i^{(1)}, x_i^{(2)}, ..., x_i^{(T_i)}\}$ denote the longitudinal vector of the i-th patient, where $x_i^{(j)}$ is denoted the j-th visit of the i-th patient, where T_i denote the number of visits for the i-th patient. In this paper, we use $\{diagnoses, procedures, medications\}$ triples to represent the patient longitudinal vector, are specified by $x_i^{(j)} = \{d_i^{(j)}, p_i^{(j)}, m_i^{(j)}\}$, where $d_i^{(j)}$ is the j-th diagnostic record of the i-th patient, and $p_i^{(j)}$ and $m_i^{(j)}$ are similarly interpreted. Let D, P, M denote the set of elements of the patient, where $D_i = \{d_i^{(1)}, d_i^{(2)}, ..., d_i^{(T_i)}\}$ denote the set of diagnoses of the i-th patient, likewise $P_i = \{p_i^{(1)}, p_i^{(2)}, ..., p_i^{(T_i)}\}$ denote the set of procedures of the i-th patient, and use $M_i = \{m_i^{(1)}, m_i^{(2)}, ..., m_i^{(T_i)}\}$ denote the set of medications of the i-th patient.

Definition 2 *(Safe Medication Recommendation)*. The goal of safe medication recommendations is to recommend safer (lower DDI) medication combinations for patients while ensuring accuracy. We train the model by the patient historical diagnoses data D_i, the patient historical procedures data P_i, and the patient historical medication data M_i, which the model can recommend a medication combination $\hat{M}_i$ for the i-th patient, where $\hat{M}_i = \{\hat{m}_1, \hat{m}_2, ..., \hat{m}_{|M|}\}$, $\hat{m}_j \in \{0, 1\}^{|M|}$ represents whether the model recommends the j-th drug.

The notations used in the PIMNet model are described in Table 1.

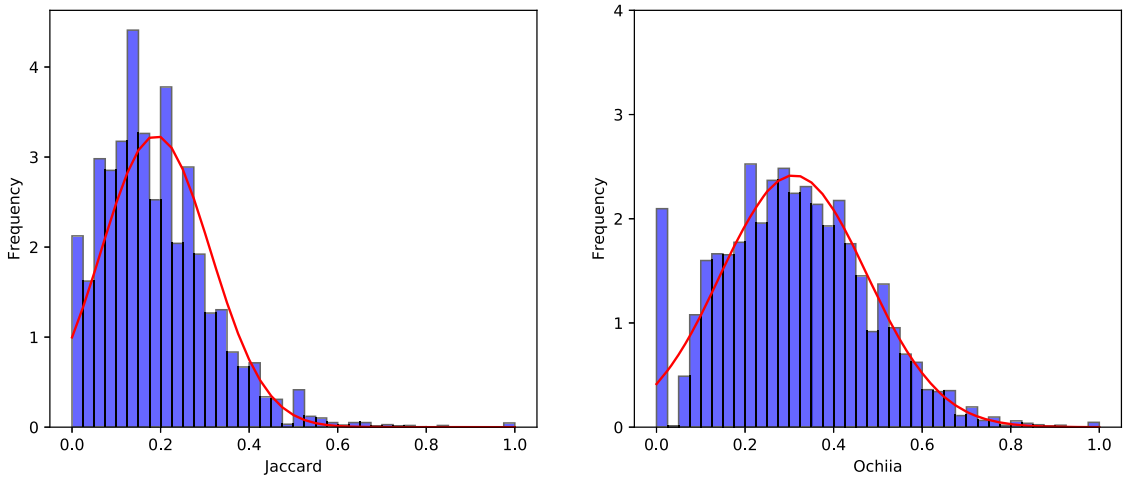


Fig. 2. Histograms of Jaccard coefficients and Ochiia coefficients between diagnoses for each patient visit in the MIMIC-III dataset.

Table 1
Description of notations used in PIMNet.

Notation	Description
X_i	the longitudinal vector of patient i
T	number of patient visits
G	the molecular graph set
D, P, M	diagnoses, procedures and medication set
$h_i^{(j)}$	the j-th patient representation of patient i
$e_i^{(j)}$	the j-th GRU output of patient i
$a_i^{(j)}$	the j-th attention score of patient i
q_i	the core condition vector of patient i
q_i'	the holistic medication vector of patient i
$\hat{\mathbf{o}}_i$	the model output of patient i
$\hat{M}_i$	recommended medication combination of patient i
M_i	the ground truth of medication combination

3.2. Overall model architecture

The model process of PIMNet is shown in Fig. 3. The whole model consists of three modules: the first module is the patient information mining module, which includes Condition Extractor Layer and Condition Evolving Layer, the task of this module is to form the current patient's core condition vector according to the patient's historical visit data. The patient information mining module improves model safety by effectively reducing DDI; the second module is the drug graph embedding module, which is to use the drug molecular graph structure to obtain the embedding of each drug; the third module is the holistic medication module, this module is designed to expand the patient's medication recommendation combination to obtain a medication expansion vector. The accuracy of the model can be improved by using the holistic medication module.

3.3. Patient information mining module

The patient information mining module aims to obtain the core condition information of patients through patient history data, reducing the DDI rate of recommended medications. This module consists of two layers:

Condition Extractor Layer

First, splice the process vector $p_i^{(t)} \in P_i$ and diagnostic vector $d_i^{(t)} \in D_i$ and put it into a linear embedding layer to get the patient representation $\mathbf{h}_i^{(t)}$.

$$\mathbf{h}_i^{(t)} = \sigma(W_h[p_i^{(t)} \# d_i^{(t)}] + b_h) \tag{1}$$

where $\mathbf{h}_i^{(t)}$ is the patient representation of the i-th patient at the t-th visit, # is the splice symbol, σ is the activation function of the linear embedding layer, W_h and b_h are the bias term and the weight matrix of the linear embedding layer.

For the condition extractor layer, we use GRU(Gate Recurrent Unit) to model the relationship between the patient representations, where the input to the condition extractor layer is the patient representation $\mathbf{h}_i^{(t)}$ obtained before. GRU [5] is an improvement on RNN and LSTM, which can effectively capture semantic associations between long sequences and solve the gradient vanishing problem of RNN while being faster than LSTM. The GRU formula we used is as follows:

$$u_t = \sigma\left(W^u \mathbf{h}_i^{(t)} + U^u \mathbf{e}_i^{(t-1)} + b^u\right) \tag{2}$$

$$r_t = \sigma\left(W^r \mathbf{h}_i^{(t)} + U^r \mathbf{e}_i^{(t-1)} + b^r\right) \tag{3}$$

$$\tilde{e}_i^{(t)} = \tanh(W^H \mathbf{h}_i^{(t)} + r_t \circ U^H \mathbf{e}_i^{(t-1)} + b^H) \tag{4}$$

$$\mathbf{e}_i^{(t)} = (1 - u_t) \circ \mathbf{e}_i^{(t-1)} + u_t \circ \tilde{e}_i^{(t)} \tag{5}$$

where $\mathbf{h}_i^{(t)}$ is the t-th patient representation of the i-th patient, σ is the sigmoid activation function, $\circ$ is element-wise product, $W^u, W^r, W^H \in \mathbb{R}^{n_H \times n_I}$ is the parameter matrix, $U^z, U^r, U^h \in n_H \times n_H$ also the parameter matrix, where n_H is the hidden size, and n_I is the input size.

Through the condition extractor layer, we can get the condition information $\mathbf{e}_i^{(t)}$, where $\mathbf{e}_i^{(t)}$ represents the condition information of the t-th visit of the i-th patient.

Condition Evolving Layer

A patient's condition changes over time. Similarly, if a doctor wants to prescribe a medication for a patient, the doctor must also consider the patient's disease evolution process to determine the patient's condition change trend. Therefore, similar to reality, the condition evolving layer is the core layer of the patient information mining module. The function of the condition evolving layer is to explore the evolutionary trend of the patient's condition according to the output of the condition extractor layer to obtain the core condition of patients.

First, to mine the patient's core condition, the condition information $\mathbf{e}_i^{(t)}$ output by the layer condition extractor layer will be sent to the attention module. The attention mechanism focuses limited attention on key information [19], thereby saving resources and quickly obtaining the most effective information. In this problem, attention will further extract the patient condition vector by assigning weights. The attention function we used in the condition evolving layer can be formulated as

$$\mathbf{a}_i^{(t)} = \frac{\exp\left(\mathbf{e}_i^{(t)} W \mathbf{h}_i^{(T)}\right)}{\sum_{j=1}^{T} \exp\left(\mathbf{e}_i^{(j)} W \mathbf{h}_i^{(T)}\right)} \tag{6}$$

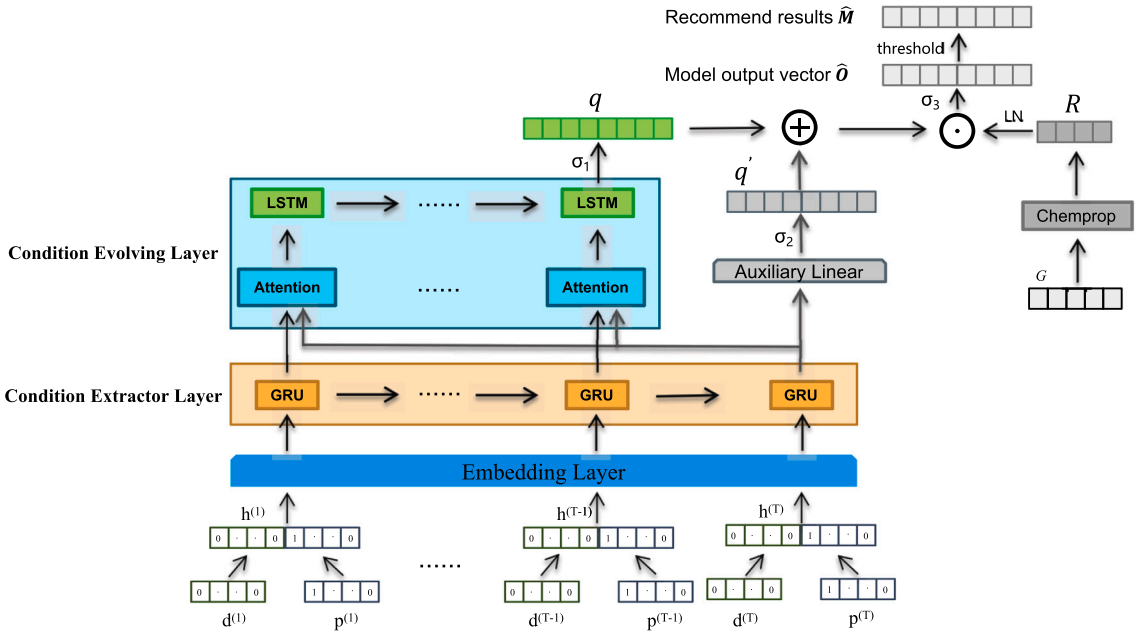


Fig. 3. The model architecture of PIMNet. For the i-th patient, we first spliced the patient's diagnoses vector $d_i^{(t)}$ and procedures vector $p_i^{(t)}$ to obtain the input $\mathbf{h}_i^{(t)}$ of the model. After the initial embedding of a layer of linear layer, input the patient representation $\mathbf{h}_i^{(1)}$ to $\mathbf{h}_i^{(T)}$, into the patient information mining module to obtain the patient's core condition vector $\mathbf{q}_i$, and input the drug molecular graph set G into Chemprop. Then add the core vector $\mathbf{q}_i$ of the patient's condition and the current patient representation $\mathbf{q}_i'$ to obtain the request vector, and multiply the request vector by the drug embedding set $\mathbf{R}$ to obtain the model output $\hat{\mathbf{o}}_i$. Finally, the model output is mapped to 0 or 1 through the threshold, and the recommended combination $\hat{\mathbf{M}}_i$ is obtained.

where $\mathbf{a}_i^{(t)}$ is the attention score of $\mathbf{e}_i^{(t)}$, W is the parameter matrix, and $\mathbf{e}_i^{(T)}$ is the current patient condition information of the i-th patient.

Using the calculated attention score $\mathbf{a}_i^{(t)}$, we can assign weights to each patient condition information $\mathbf{e}_i^{(t)}$ to highlight the historical condition information that has a greater effect on the patient's current visit. We use the attention score $\mathbf{a}_i^{(t)}$ multiplied by the output $\mathbf{e}_i^{(t)}$ of the condition extractor layer as the input $\mathbf{x}_i^{(t)}$ of the LSTM module, as follows:

$$\mathbf{x}_i^{(t)} = \mathbf{a}_i^{(t)} \times \mathbf{e}_i^{(t)} \tag{7}$$

In the last part of the condition evolving layer, we use LSTM [16] to simulate the evolution of the patient's condition. The formula of LSTM used in PIMNet is as follows:

$$i^t = \sigma\left(\left(W_{xi}\mathbf{x}_i^{(t)} + W_{qi}\mathbf{q}_i^{(t-1)} + W_{ci} \circ c^{t-1} + b_i\right)\right) \tag{8}$$

$$f^t = \sigma\left(W_{xf}\mathbf{x}_i^{(t)} + W_{qf}\mathbf{q}_i^{(t-1)} + W_{cf} \circ c^{t-1} + b_f\right) \tag{9}$$

$$c^t = f^t \circ c^{t-1} + i^t + b_c \tag{10}$$

$$o^t = \sigma\left(W_{xo}\mathbf{x}_i^{(t)} + W_{qo}\mathbf{q}_i^{(t-1)} + W_{co} \circ c^t + b_o\right) \tag{11}$$

$$\mathbf{q}_i^{(t)} = o^t \circ \tanh\left(c^t\right) \tag{12}$$

where W is the parameter matrix, $\mathbf{x}_i^{(t)}$ is the patient feature vector at time t, and c^t is the memory at time t. As the hidden state of an LSTM cell, $\mathbf{q}_i^{(t)}$ will be passed down.

We put the last hidden state vector $\mathbf{q}_i^{(T-1)}$ into a fully connected layer to obtain the core condition vector $\mathbf{q}_i$.

$$\mathbf{q}_i = \sigma_1(W_q\mathbf{q}_i^{(T-1)} + b_q) \tag{13}$$

where σ_1 is the activation function of the fully-connected layer, W_q and b_q are the bias term and the weight matrix of the fully connected layer.

Through the condition extractor layer and condition evolving layer, we get the patient's core condition vector $\mathbf{q}_i$ of the i-th patient at the current moment.

3.4. Drug graph embedding module

The task of the medication graph encoder is to encode a graph of medication molecules and obtain the medication embedding. Inspired by Gilmer et al. [7] and Stokes et al. [17], we use the directed message passing neural network (Chemprop) as the medication graph encoder in this paper.

Chemprop can be considered as a variant of the general MPNN [7]. The most notable difference between them is the nature of the messages sent during the messaging phase.

We first collected the set of drugs M used in the dataset. According to the drug set M, we can get the molecular image set G, the molecular graphs of these medications are composed of atoms and atom-atom edges. From the molecular image of the i-th drug G_i, the neighbor set $N(v)$ of the v node in the graph G_i can be obtained. We let a_v represent the node v (atom v) features, e_{vw} represents the edge features (bond) between node v and node w. The calculation process of Chemprop is as follows:

$$m_{vw}^{t+1} = \sum_{k \in \{N(v) \setminus w\}} MESSAGE_t\left(a_v, a_k, h_{kv}^t\right) \tag{14}$$

$$h_{vw}^{t+1} = UPDATE_t\left(h_{vw}^t, m_{vw}^{t+1}\right) \tag{15}$$

where t is the layer number, h_{kv}^t is hidden states between node k and node v, m_{vw}^t is the encoded messages between node v and node w at the t-th iteration, W_i is a learned matrix, $cat\left(a_v, e_{vw}\right)$ is the concatenation of the atom features a_v for atom v and the bond features e_{vw} between node v and w. $MESSAGE_t()$ is used as a message function to pass messages between vertices, and $UPDATE_t()$ is an update function used to update vertices.

$$UPDATE_t\left(h_{vw}^t, m_{vw}^{t+1}\right) = \tau\left(h_{vw}^0 + W_m m_{vw}^{t+1}\right) \tag{16}$$

$$h_{vw}^0 = \tau\left(W_i \operatorname{cat}\left(a_v, e_{vw}\right)\right) \tag{17}$$

The calculation formula of the vertex update function $UPDATE_t()$ and the definition of the initial hidden state h_{vw}^0 are as above. Finally, we add all hidden states h_{kv}^t to get the molecular graph embedding

$$m_v = \sum_{k \in N(v)} h_{kv}^T \tag{18}$$

$$h_v = \tau\left(W_a \operatorname{cat}\left(a_v, m_v\right)\right) \tag{19}$$

$$\mathbf{r}_i = \sum_{v \in G} h_v \tag{20}$$

where $W_a \in \mathbb{R}^{h \times h}$ is a learned matrix, h_v is the embedding of node v (atom v), $\mathbf{r}_i$ is the embedding of graph G_i (molecular image of drug M_i). We collect all drug embeddings into the drug embedding set $\mathbf{R}$ as the drug information required by the model.

3.5. Holistic medication module

After going through the patient information mining module, we can get the patient's core condition vector, and the medication recommendation for this vector can greatly reduce the DDI of the recommended combination. However, the medication recommendation through the core disease vector will only get the core medications of the recommended combination, which leads to a smaller number of drugs in the recommended results and thus affects the model accuracy. Therefore, the holistic medication module is needed to supplement the recommended combination. In reality, when doctors prescribe medications to patients, they will give core medications based on the patient's main disease, and then give refined medication combinations based on the patient's condition.

The holistic medication module will only recommend based on the current disease information $\mathbf{e}_i^{(T)}$. This module inputs $\mathbf{e}_i^{(T)}$ into the auxiliary linear layer, and the input obtained through the holistic medication linear layer is the current medication vector $\mathbf{q}_i'$.

$$\mathbf{q}_i' = \sigma_2(W_p \mathbf{e}_i^{(T)} + b_p) \tag{21}$$

where $\mathbf{q}_i'$ denotes the holistic medication vector of the i-th patient at the current moment, σ_2 is the activation function of the auxiliary linear layer, W_p and b_p are the bias term and the weight matrix of the auxiliary linear layer.

Then add the patient's core condition vector $\mathbf{q}_i$ and the holistic medication vector $\mathbf{q}_i'$ to get the request vector, and then multiply the request vector with the drug embedding set R obtained in the drug graph embedding module before getting the final result, the formula is as follows:

$$\hat{\mathbf{o}}_i = \sigma_3\left(\left(\mathbf{q}_i{}' + \lambda \mathbf{q}_i\right) \odot \mathbf{R}\right) \tag{22}$$

where λ is a learnable parameter, which value is automatically adjusted according to the model training results, $\odot$ is element-wise multiplication, $\hat{\mathbf{o}}_i$ is the model output for the i-th patient.

Finally, we convert the output of the model into a classification label, for each value in $\hat{\mathbf{o}}_i$, when the value is greater than 0.5, it is assigned a value of 1, otherwise, it is assigned a value of 0, and the obtained $\hat{\mathbf{M}}_i$ is the recommended medication combination, the length of $\hat{\mathbf{M}}_i$ is the number of the medication.

3.6. Model training

In [23], it has been mentioned in previous work that the ground truth DDI rate is 0.0808. Therefore, for the drug recommendation model, we must consider the interaction between the accuracy of the recommendation results and the DDI rate when designing the loss function, because too low DDI rate may lead to a decrease in the accuracy of the results.

In this paper, we use the binary cross-entropy loss function and the multi-label hinge loss function as the measure of the accuracy of the recommendation result, and the DDI function as the measure of the recommendation result DDI, and finally give the loss function of this paper comprehensively.

Binary Cross-Entropy Loss. In classification tasks, cross-entropy is often used to measure the difference between target and output, the formula for BCE is as follows:

$$L_{bce} = -\sum_{i=1}^{|M|} \mathbf{m}_i^{(t)} \log\left(\hat{\mathbf{o}}_i^{(t)}\right) + \left(1 - \mathbf{m}_i^{(t)}\right) \log\left(1 - \hat{\mathbf{o}}_i^{(t)}\right) \tag{23}$$

where $\mathbf{m}_i^{(t)}$ is the prediction target, $\hat{\mathbf{o}}_i^{(t)}$ represents the t-th element (the t-th drug) of the recommendation vector for the i-th patient and $|M|$ is the length of medication set.

Multi-label Hinge Loss. Multi-label hinge loss considers the error between the corresponding real category and other categories, and is often used to measure the accuracy of multi-label classification. The formula is as follows:

$$L_{\text{multi}} = \sum_{i,j:\mathbf{m}_i^{(t)}=1,\mathbf{m}_j^{(t)}=0} \frac{\max\left(0, 1 - \left(\hat{\mathbf{o}}_i^{(t)} - \hat{\mathbf{o}}_j^{(t)}\right)\right)}{|M|} \tag{24}$$

where $\hat{\mathbf{o}}_i^{(t)}$ represents the t-th element (the t-th drug) of the recommendation vector for the i-th patient and $|M|$ is the length of medication set.

DDI Loss. According to the DDI data, we can calculate the DDI of the model recommendation combination, we design the following formula to calculate the DDI loss:

$$L_{ddi} = \sum_{i=1} \sum_{j=1} \mathbf{I}_{ij} \cdot \hat{\mathbf{o}}_i^{(t)} \cdot \hat{\mathbf{o}}_j^{(t)} \tag{25}$$

where I is the matrix data of DDI, extracted by [18]. I_{ij} is used to indicate whether there is a drug-drug interaction between drugs i and j, if so, I_{ij} is 1 otherwise I_{ij} is 0.

Loss Function. To synthesize the above three loss functions, we use pre-defined parameters to train with multiple loss functions [6]. The loss function we designed is as follows:

We give a DDI threshold μ, which is set by the requirements of the task. In the experiment, we set μ to 0.05, when the DDI does not exceed μ, we think this DDI is acceptable, and the task of the loss function is to improve model accuracy. The formula of the loss function at this time is as follows:

$$L = \left(\alpha L_{bce} + (1 - \alpha) L_{multi}\right) \tag{26}$$

when the DDI of the model result exceeds the threshold μ, the loss function will comprehensively consider both the accuracy and the DDI. At this time, the loss formula is as follows:

$$L = \left(\alpha L_{bce} + (1 - \alpha) L_{multi}\right) + \beta L_{ddi} \tag{27}$$

4. Experiment

4.1. Dataset

Our experiments are based on the public dataset MIMIC-III raised by [10], which contains nearly 60,000 ICU admission records. The statistics of the post-processed data are reported in Table 2. In the experiment, we take two-thirds of the dataset as the training set, one-sixth as the validation set, and one-sixth as the test set.

4.2. Baseline

In our paper, we evaluate the performance of PIMNet by comparing it with the following baseline methods:

- **LR:** standard Logistic Regression;
- **ECC** [14]**:** Ensemble Classifier Chain (ECC)
- **RETRIN** [4]**:** which uses two sets of attention mechanisms to make recommendations by simulating a diagnosis process is the first model to use patient historical data;
- **LEAP** [24]**:** which base on attention mechanism and introducing external medical knowledge to improve return results;
- **DMNC** [13]**:** which prediction via memory augmented neural network based on differentiable neural computers;

Table 2
Data Statistics of MIMIC-III.

item	Number
Number of patients	6350
Number of diagnosis	1958
Number of procedures	1430
Number of medications	151
Average number of visits per patient	2.36
Average number of diagnosis per visit	13.63
Average number of procedures per visit	4.53
Average number of medications per visit	19.85

Table 3
The best Performance Comparison on MIMIC-III.

Model	DDI	Jaccard	F1	PRAUC
LR	0.0829	0.4865	0.6434	0.7509
ECC	0.0846	0.4996	0.6569	0.6844
RETAIN	0.0835	0.4887	0.6481	0.7556
LEAP	0.0731	0.4521	0.6138	0.6549
DMNC	0.0842	0.4864	0.6529	0.7580
GAMENet	0.0864	0.5067	0.6626	0.7631
MICRON	0.0641	0.5100	0.6654	0.7687
SafeDrug	0.0589	0.5213	0.6768	0.7647
COGNet	0.0852	0.5336	0.6869	**0.7739**
PIMNet(Ours)	**0.0481**	**0.5382**	**0.6883**	0.7655

Table 4
Ablation Study for PIMNet on MIMIC-III.

Model	DDI	Jaccard	F1	PRAUC
PIMNet $w/o\ q_i'$	**0.0383**	0.4736	0.6211	0.7544
PIMNet $w/o\ q_i$	0.0786	0.5201	0.6783	0.7556
PIMNet $w/o\ D$	0.0502	0.4733	0.6232	0.7456
PIMNet $w/o\ P$	0.0516	0.4912	0.6502	0.7606
PIMNet	0.0481	**0.5382**	**0.6883**	**0.7655**

- **GAMENet** [15]**:** which introduces EHR graph and DDI graph through GCN method, and uses graph memory module for recommendation;
- **MICRON** [22]**:** which makes recommendations by modeling prescription changes;
- **SafeDrug** [23]**:** which makes medication recommendation by introducing molecular structure diagram;
- **COGNet** [21]**:** which introduces a new prediction mechanism for recommendation, and also introduces medication vectors and EHR

4.3. Performance of PIMNet on the dataset

As shown in Table 3, the PIMNet proposed in this paper greatly reduces the DDI of the recommended medication combination, while outperforming the state-of-the-art methods on the two metrics of Jaccard and F1-score.

For methods that do not consider patient history information, namely LR, ECC raised by [14] and LEAP raised by [4], these method values are recommended based on current patient health information, which leads to poor results for these methods. For methods that consider patient history information, namely RETAIN raised by [4], DMNC raised by [13], GAMENet raised by [15], SafeDrug raised by [23], MICRON raised by [22] and COGNet raised by [21], the results of these methods are better. SafeDrug further improves the performance by introducing the molecular structure of the medication, and its proposed Local Bipartite Encoder greatly reduces the DDI of the recommended medication combination. The MICRON model was the first to propose a method for modeling changes in drugs, however its results were slightly lower due to the lack of modeling of disease relationships and the relationship between medications and disease. COGNet uses a transformer-like [19] structure to model medications and introduces medication vectors and patient electrical health records graphs, but the DDI of its recommended results is high.

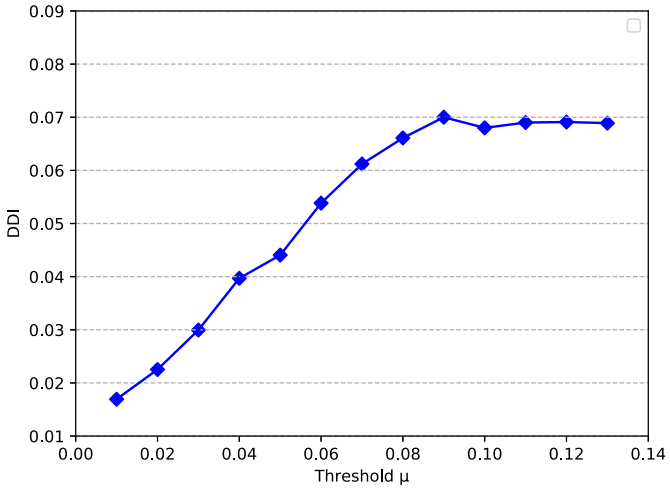


Fig. 4. DDI results obtained by gradually adjusting the DDI threshold μ.

4.4. Parameter experiment of threshold μ

In the PIMNet, the parameters that need to be manually adjusted are mainly the DDI threshold μ in the loss function, which is used to control the maximum DDI rate that the model recommendation result can achieve. To investigate the effect of this parameter on the DDI, we gradually adjusted the value of the parameter for training and plotted the results as a function of the parameter in Fig. 4.

We gradually adjust the DDI threshold μ to obtain the performance of the model under different thresholds. Specifically, we gradually increase the DDI threshold μ, keeping other parameters constant to train the model. In Fig. 4, it can be found that when the threshold μ is less than 0.05, the DDI rate increases with the increase of μ. When the threshold continues to increase, the DDI rate of the model is stable at around 0.07.

According to the experimental results, we can draw the following conclusions:

- When the threshold μ is less than 0.06, the threshold μ has a great inhibitory effect on DDI;
- The maximum value of the model DDI rate is around 0.07, which also indirectly proves the security of PIMNet.

4.5. Ablation study

To verify the effectiveness of each module in the PIMNet model, we designed the following ablation studies:

- PIMNet $w/o\ q_i'$ We remove the holistic medication module, just use the patient's core condition vector q_i.
- PIMNet $w/o\ q_i$ We remove the patient information mining module, just use the holistic medication vector q_i'.
- PIMNet $w/o\ D$ We remove the diagnoses information in each visit.
- PIMNet $w/o\ P$ We remove the procedures information in each visit.

The results of each module ablation experiment are listed in Table 4.

For the ablation experiment results of q_i', it can be seen that the patient information mining module can effectively extract the core condition of the patient, and achieve lower DDI in the case of less recommended medications. However, when only use the patient's core

Table 5
An example of case study.

Order of visits	Diagnoses	Recommended Medication Combination
1st visit	Cirrhosis of liver Congestive heart failure Pancytopenia	**correct**: B05C, A12C, A07A, N01A, A06A, N02B, C03C, A12A, A02B, C01C, N03A, C02D, N05C, R01A, J01D, C01B **missed**: C07A, H01C, R03A **error**: D07A
2nd visit	Hepatic coma Acute respiratory failure Congestive heart failure Iron deficiency anemia Pneumonia	**correct**: N02B, A06A, B05C, A12A, A12C, C01C, A07A, N01A, C07A, A12B, C03C, A02A, B01A, A02B, C01B, A01A, H03A, N03A, N05C, C09A, A03F, A05A, C05A **missed**: J01M **error**: J01C, A01A
3rd visit	Acute renal failure Congestive heart failure Pneumonia	**correct**: A01A, A02B, A12C, C07A, C10A, A12B, C03C, A12A, B05C, B01A, N03A, A06A, H03A, N05C, A04A, C09A, R01A, R03A, J01X, J01C, **missed**: A07A

condition vector q_i, the number of drugs in the recommended combination given by the model is small, which make lower Jaccard similarity coefficients and F1 score.

For the ablation experiment results of q_i, it can be seen that the model can also give acceptable results using only the holistic medication module, but the lack of the patient's core condition vector q_i makes the model DDI rate higher.

For the ablation experiments of D and P, the experimental results are pool due to the lack of D or P, which proves the necessity of entering D and P.

The ablation experiments on the above problems are all lower than PIMNet, which proves the effectiveness of PIMNet and related modules, which also proves that PIMNet can effectively reduce DDI and provide safer medication recommendations.

4.6. Case study

To illustrate the output of the model, we select a patient from the test dataset for analysis. As shown in Table 5, where **correct** refers to that the model results are consistent with the ground truth, **missed** refers to the drugs that the model failed to recommend compared with the ground truth, and **error** refers to the additional results compared to the ground truth.

It can be found that when the model recommends the patient's first visit, due to the lack of the patient's historical condition information, the model has some errors. As the number of patient visits increased, the results of the model gradually improved, although the patient developed a new condition. From the perspective of drugs, some drugs that appeared last time, such as A12A and N01A, have been correctly recommended, and the model's ability to recommend new drugs is gradually increasing.

5. Conclusion

In this paper, we propose a medication recommendation model, PIMNet, which can better capture the evolution of a patient's condition by using the patient's historical visit information, thereby giving a more accurate and smaller DDI medication combination. We verified the performance of the model on the public dataset MIMIC-III. The results show that PIMNet can greatly reduce the DDI rate of the recommended combination under the condition of providing accurate medication recommendation combinations. In fact, the purpose proposed by PIMNet is to reduce the DDI rate of the recommended combination. In some actual treatment cases, the low DDI rate is even more important than the accuracy of the medication combination itself. At the end of the article, we demonstrate the effectiveness of each component in PIMNet through ablation experiments and parametric experiments.

In future work, continuing to mine new data in the dataset may further improve model performance. We will continue to find new ways to reduce the recommended combination DDI rate by further mining the patient's condition information.

CRediT authorship contribution statement

Ruobing Li: Conceptualization, Formal analysis, Investigation, Methodology, Software, Writing – original draft. **Jian Wang:** Data curation, Writing – original draft. **Hongfei Lin:** Investigation, Visualization. **Yuan Lin:** Resources, Supervision. **Huiyi Lu:** Validation. **Zhihao Yang:** Visualization, Writing – review & editing.

Declaration of competing interest

- The authors declare that there are no conflict of interests, we do not have any possible conflicts of interest.
- The authors declare no conflict of interest.
- We declare that we have no conflict of interest.
- The authors do not have any possible conflicts of interest.
- The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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